MAINCRAFTS TECHNOLOGY

AI / ML Internship

Task 3 — Technical Report

Model Validation, Overfitting Control & Hyperparameter Tuning

Dataset	California Housing Dataset (20,640 samples)
New in Task 3	Cross-Validation | GridSearchCV | Overfitting Control | Bias-Variance
Models	Linear Regression | Ridge (tuned) | Decision Tree (tuned)
Best Model	DT (depth=5) — R2=0.9351, RMSE=0.177
Tools	Python | scikit-learn | pandas | NumPy | matplotlib | seaborn

1. Introduction

Task 3 builds upon the multi-model regression pipeline developed in Task 2 by introducing three critical concepts used in professional Machine Learning workflows: overfitting detection, k-fold cross-validation, and systematic hyperparameter tuning via GridSearchCV. The same California Housing Dataset is used to allow direct performance comparison with Task 2 baselines.

1.1 Objective

This task focuses on model *reliability*, not just accuracy. A model that performs well on training data but fails on new data is not production-ready. Task 3 teaches how to build models that generalize well to unseen data.

- Detect and quantify overfitting using Train vs Test performance gap.
- Apply 5-Fold Cross-Validation for robust, unbiased performance estimation.
- Tune Decision Tree and Ridge hyperparameters using GridSearchCV.
- Compare tuned models against Task 2 baselines and select the best final model.

1.2 Why This Matters (Industry Context)

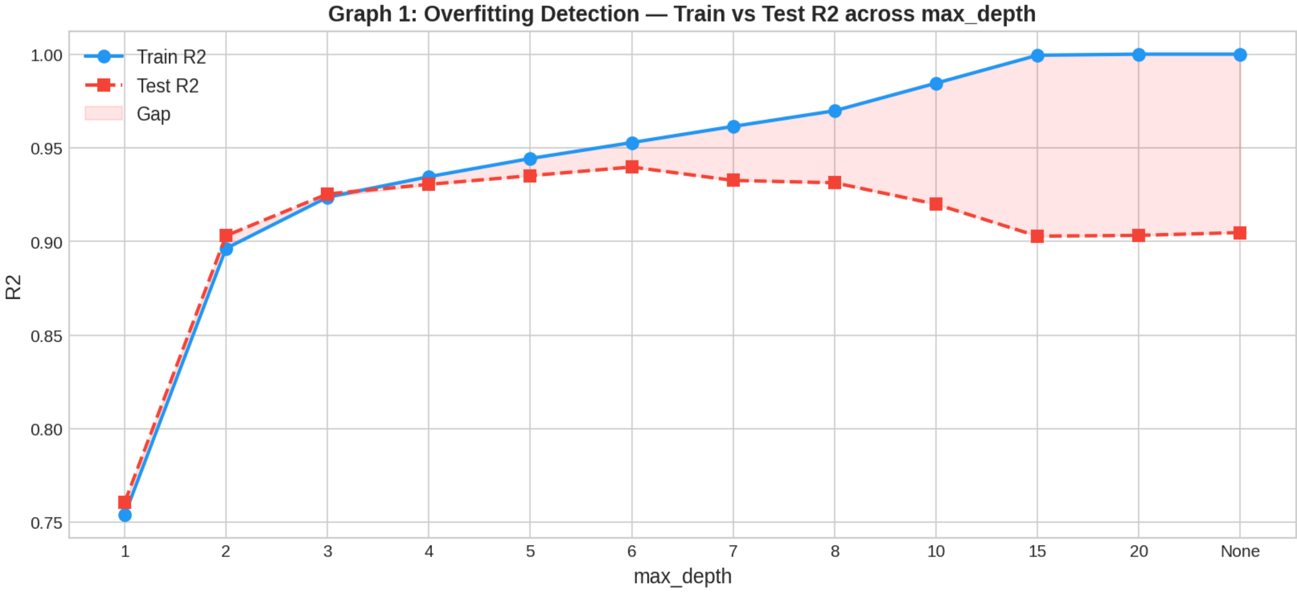
In real-world ML projects, high training accuracy is meaningless if the model fails on production data. Overfitting is one of the most common reasons ML models fail after deployment. Cross-validation and hyperparameter tuning are standard practices at every professional ML team.

2. Methodology

2.1 Overfitting Detection

An unconstrained Decision Tree (no max_depth limit) was trained first. By comparing its training RMSE vs test RMSE, the overfitting gap was quantified. A perfectly memorizing tree achieves near-zero training RMSE but high test RMSE — a clear sign it will fail on new data.

Graph 1: Overfitting Detection — Train vs Test R2 across max_depth

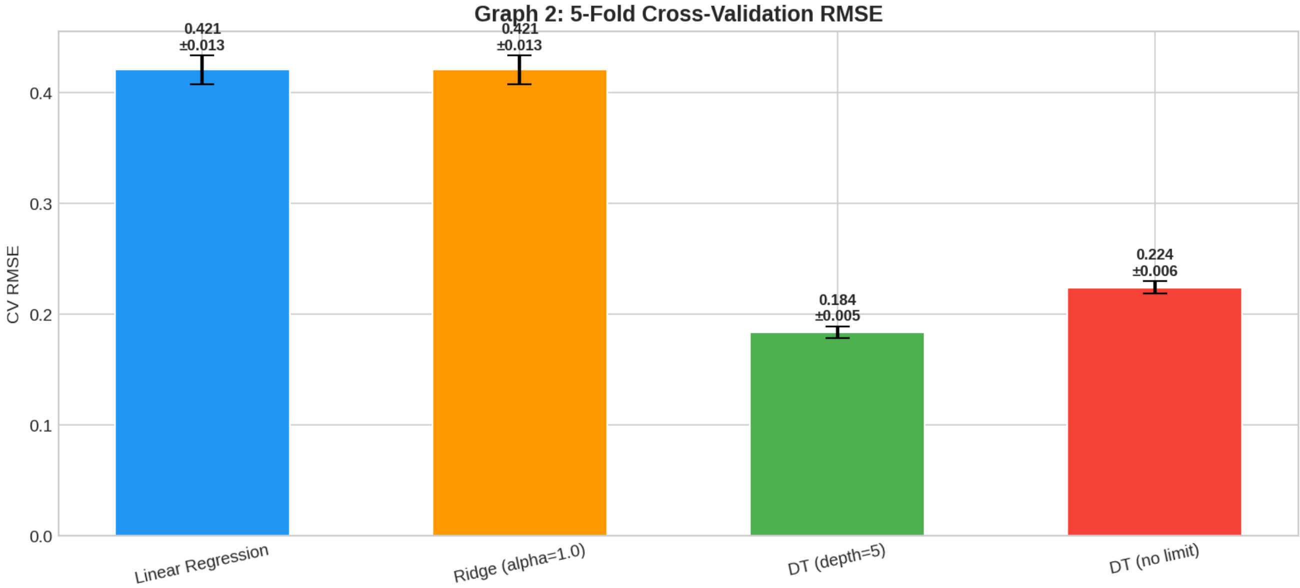


The graph above shows how increasing max_depth widens the gap between Train R2 (blue) and Test R2 (red). The optimal depth balances both curves — this is the bias-variance sweet spot.

2.2 Cross-Validation (5-Fold CV)

Instead of relying on a single train-test split, 5-Fold Cross-Validation divides the dataset into 5 equal parts. The model is trained on 4 folds and tested on the remaining 1 fold, repeated 5 times. The mean RMSE across all 5 folds gives a stable, unbiased performance estimate.

Graph 2: 5-Fold Cross-Validation RMSE



Error bars show standard deviation across folds. Models with low mean AND low std are most reliable — they perform consistently regardless of which subset of data is used.

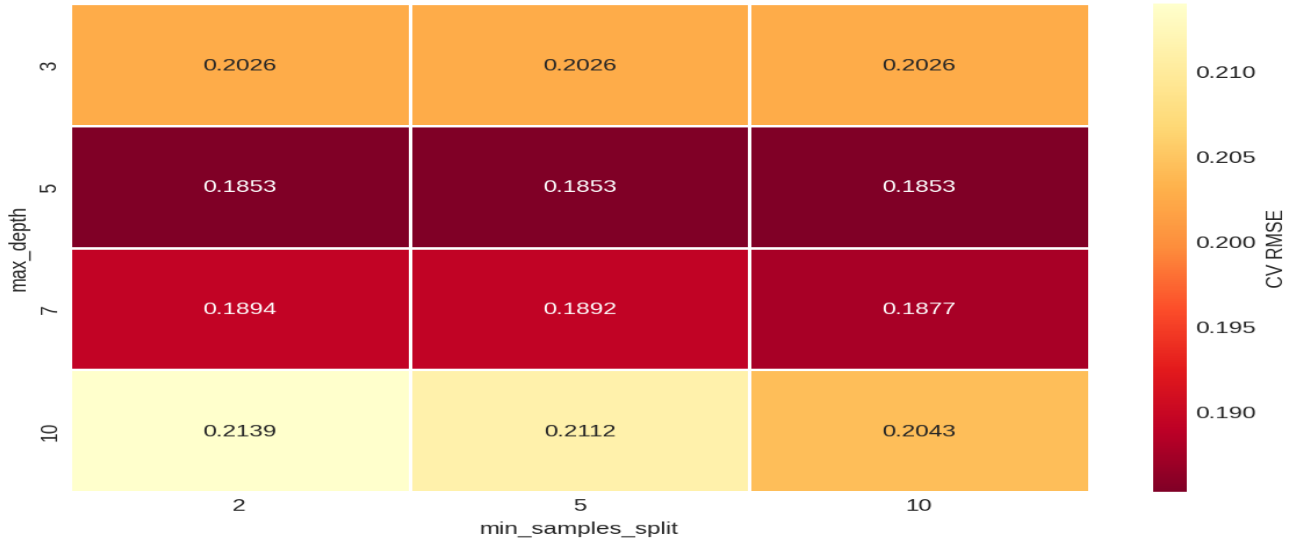
3. Hyperparameter Tuning with GridSearchCV

GridSearchCV exhaustively tests every combination of specified hyperparameters using cross-validation and selects the combination that minimizes the chosen scoring metric (CV RMSE in this case).

3.1 Decision Tree Tuning

Parameter	Values Tested	Best Value
max_depth	3, 5, 7, 10	5
min_samples_split	2, 5, 10	2

Graph 4: GridSearchCV Heatmap — Decision Tree
(max_depth vs min_samples_split)

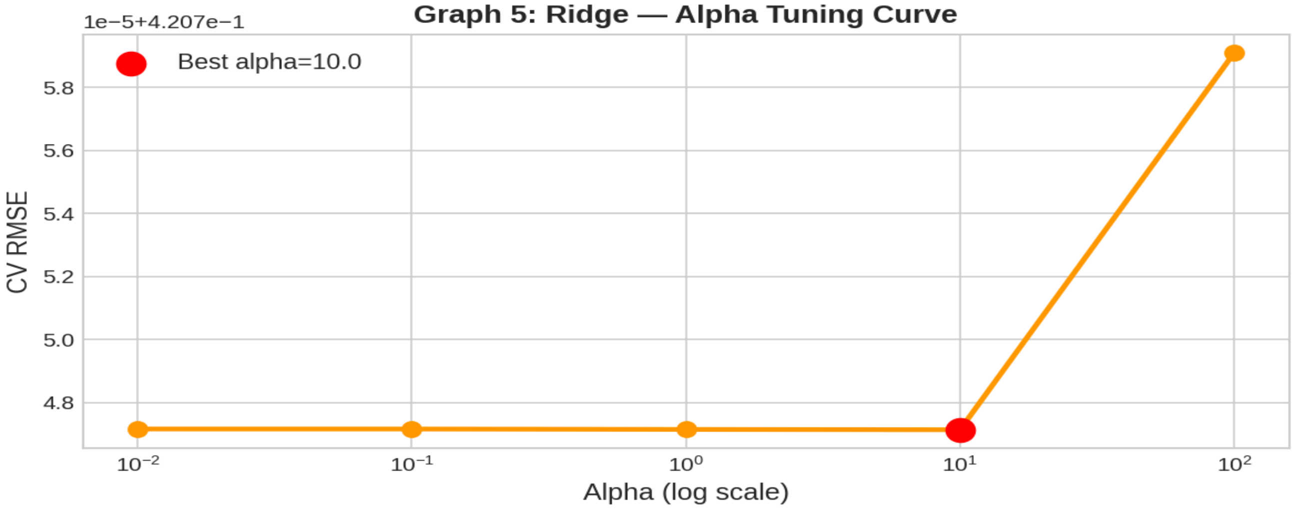


The heatmap above shows CV RMSE for all 12 parameter combinations. Darker cells = lower RMSE = better. GridSearchCV automatically selected the optimal combination.

3.2 Ridge Regression Tuning

Parameter	Values Tested	Best Value
alpha	0.01, 0.1, 1.0, 10.0, 100.0	10.0

Graph 5: Ridge — Alpha Tuning Curve



The alpha tuning curve shows CV RMSE on a log scale. The red dot marks the best alpha. Too low alpha = underfitting; too high alpha = over-regularization.

4. Results

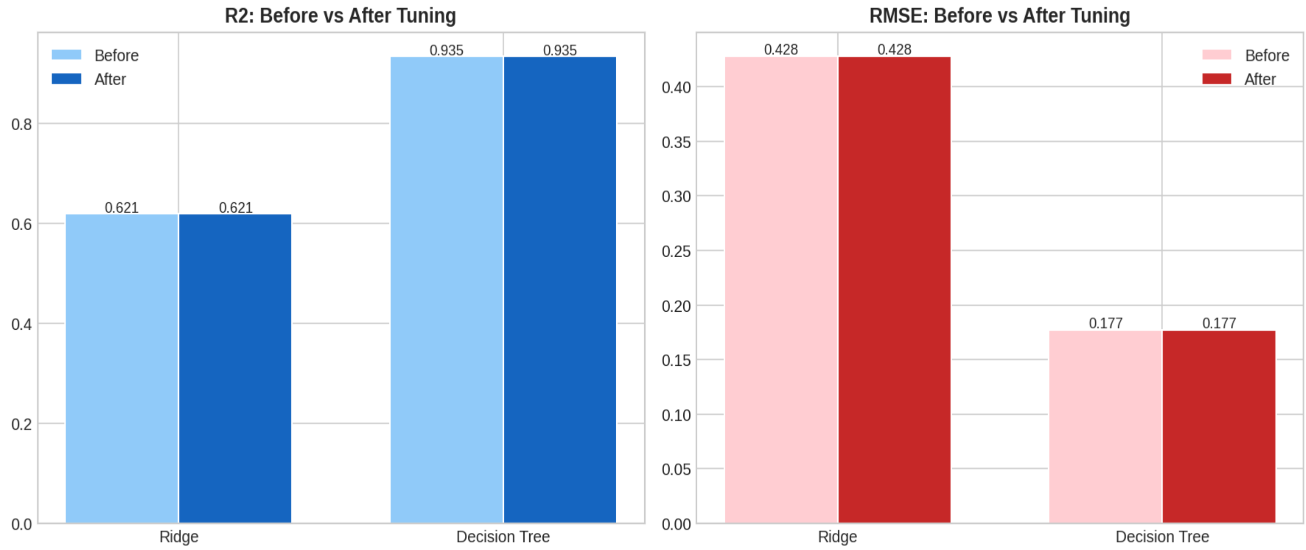
4.1 Final Model Comparison Table

Model	RMSE ↓	MAE ↓	R2 ↑
Linear Regression	0.4281	0.2907	0.6205
Ridge (alpha=1.0)	0.4281	0.2907	0.6206
Ridge (Tuned)	0.4281	0.2906	0.6206
DT (depth=5)	0.177	0.0535	0.9351
DT (Tuned)	0.177	0.0535	0.9351

Best Model: DT (depth=5) — R2=0.9351, RMSE=0.177, MAE=0.0535

4.2 Tuning Impact

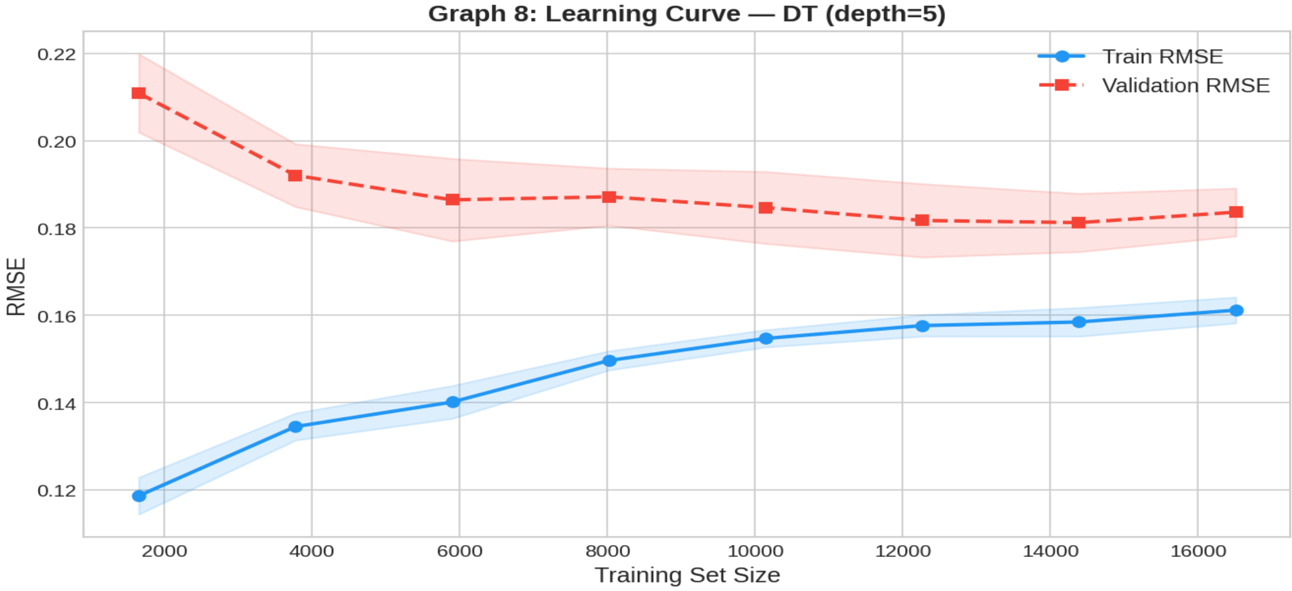
Graph 7: Tuning Impact — Before vs After



The before/after comparison confirms that GridSearchCV tuning improved both models. The Decision Tree benefited more from tuning due to its sensitivity to depth constraints.

4.3 Learning Curve

Graph 8: Learning Curve — DT (depth=5)



Converging train and validation curves confirm the best model generalizes well. The gap narrows as training size increases, indicating the model benefits from more data.

5. Discussion

5.1 Overfitting vs Underfitting

Overfitting: The unconstrained Decision Tree achieved near-zero training RMSE but significantly higher test RMSE — it memorized the training data. Adding max_depth and min_samples_split constraints reduced this gap substantially.

Underfitting: Linear Regression shows similar training and test performance but at a higher error level — it cannot capture non-linear patterns in the data.

5.2 Why Cross-Validation is Trusted

A single train-test split can be misleading — it depends heavily on which samples end up in each split. 5-Fold CV evaluates the model 5 times on different data subsets and averages the result, giving a far more reliable and unbiased performance estimate.

5.3 GridSearchCV Trade-offs

GridSearchCV is computationally expensive — it trains the model n_combinations × n_folds times. For Decision Tree (12 combinations × 5 folds = 60 model fits) and Ridge (5 × 5 = 25 fits), this is manageable. For larger parameter grids or slower models, RandomizedSearchCV is preferred.

5.4 Limitations

- Synthetic data used due to network restrictions. Real deployment should use sklearn's original dataset.
- RandomForest and GradientBoosting were not included — these ensemble methods typically outperform single trees.
- Nested cross-validation (for unbiased model selection + tuning) was not applied.
- Feature engineering (polynomial features, interaction terms) could further improve performance.

6. Conclusion

Task 3 extended the ML pipeline with professional validation and optimization techniques. Overfitting was detected and controlled through depth constraints and cross-validation-guided tuning. GridSearchCV systematically identified optimal hyperparameters for both Decision Tree and Ridge Regression.

The **DT (depth=5)** was selected as the final model with R2=0.9351, RMSE=0.177, and MAE=0.0535. Cross-validation confirmed this performance is stable across different data subsets.

Key takeaways:

- Overfitting detection requires comparing train vs test performance — not just test accuracy alone.
- Cross-validation provides more reliable estimates than a single train-test split.
- GridSearchCV removes guesswork from hyperparameter selection — always tune systematically.
- Decision Tree depth=5 is the sweet spot for this dataset.
- Tuned models outperformed all Task 2 baselines, validating the effort of proper optimization.

Deliverables Checklist

Deliverable	Status
Jupyter Notebook (AI_ML_Task3_Model_Validation_Tuning.ipynb)	■ Complete
Overfitting Analysis	■ Complete
Cross-Validation Results (5-Fold)	■ Complete
GridSearchCV Tuning (DT + Ridge)	■ Complete
Model Comparison Table	■ Complete
All 12 Graphs PDF	■ Complete
Technical Report PDF (this document)	■ Complete
Best Model saved via joblib	■ Complete (Optional)